

HDOC DAY-15

Menu-Driven C Program Using Switch-Case

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Batch	15

1. Problem Statement

Prepare the algorithm and test case table. Then write, compile, execute, and test a C program using switch-case. The program accepts a positive integer and performs one operation according to the user's choice:

1. Check whether all digits are even.
2. Check whether all digits are prime (prime digits are 2, 3, 5 and 7).

2. Algorithm

1. Start.
2. Input a positive integer n.
3. Display the menu:
 - 1 - Check whether all digits are even
 - 2 - Check whether all digits are prime
4. Input the user's choice.
5. Use switch(choice).
6. For choice 1:
 - Examine every digit using $\text{digit} = n \% 10$.
 - If any digit is odd, mark the condition false.
 - Remove the last digit using $n = n / 10$.
 - Print whether all digits are even.
7. For choice 2:
 - Examine every digit.
 - If a digit is not 2, 3, 5 or 7, mark the condition false.
 - Print whether all digits are prime.
8. For any other choice, print "Invalid choice".
9. Stop.

3. Test Case Table

Test	Number	Choice	Expected Output
1	12345	1	All digits are not even
2	24680	1	All digits are even
3	2357	2	All digits are prime
4	2379	2	All digits are not prime

4. C Program

```
#include <stdio.h>

int main() {
    int n, temp, digit, choice;
    int allEven = 1, allPrime = 1;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    printf("\nMENU\n");
    printf("1. Check whether all digits are even\n");
    printf("2. Check whether all digits are prime\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    temp = n;

    switch (choice) {

        case 1:
            while (temp > 0) {
                digit = temp % 10;

                if (digit % 2 != 0) {
                    allEven = 0;
                    break;
                }
            }

            if (allEven)
                printf("All digits are even\n");
            else
                printf("All digits are not even\n");

            break;

        case 2:
            while (temp > 0) {
                digit = temp % 10;

                if (!(digit == 2 || digit == 3 ||
                    digit == 5 || digit == 7)) {
                    allPrime = 0;
                    break;
                }

                temp = temp / 10;
            }

            if (allPrime)
                printf("All digits are prime\n");
            else
                printf("All digits are not prime\n");

            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

5. Compilation, Execution and Testing

The program was compiled and executed for the prepared test cases. The terminal examples below use the requested Mac terminal name **gurveer@Gurveers-MacBook-Air**.

```
Last login: Sat Sep  5 15:47:37 on ttys000
Enter a positive integer: 2468

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even
[vansh@vanshs-MacBook-Air-3 c_language % ./hello
Enter a positive integer: 2357

MENU
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime
vansh@vanshs-MacBook-Air-3 c_language %
```

6. Result

The menu-driven C program using switch-case was compiled and tested successfully. It correctly checks whether all digits of the entered positive integer are even or whether all digits are prime, according to the user's selected menu option.